

Statement of Work — Real-Time Multiplayer Racing Game Cloud Backend

- **Client:** Game studio ("GC") — 15 Unity developers, 2 incoming DevOps engineers
 - **Prepared by:** Cloud consulting team ("CA")
 - **Date:** 2026-06-25
 - **Version:** 1.0 (Draft for client review)
 - **Basis:** This SOW is built entirely on the requirements and decisions in [01-requirements-and-architectural-implications.md](#). Requirement IDs (BR/FR/NFR/DR/SC) and decision IDs (D-1...D-18) are cited inline for traceability.
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1. Executive summary

GC is launching its first title — a real-time, 8-player multiplayer mobile racing game on iOS and Android — within an investor-mandated 8-month window (BR-1, BR-2). The game must support at least 100,000 concurrent players at launch in North America and Europe, expand to Asia within six months, and deliver a crisp, low-latency experience that differentiates it from laggy competitors (BR-3, BR-5, BR-7). GC's current Unity built-in networking does not scale, and the team is new to cloud operations.

This engagement delivers a scalable, secure, multi-region cloud backend on **AWS** (D-1), using managed services wherever possible so a small, cloud-new team can operate it within the timeline. Real-time match hosting runs on **Amazon GameLift** (D-2); player and game data on **DynamoDB** with a residency-aware data model (D-11); analytics on an **S3 + Athena** lake with a streaming path for cheat detection (D-5, D-9). The whole platform is defined in Terraform (D-4) on a greenfield AWS Organization landing zone (D-10), shipped through GitHub Actions CI/CD (D-13).

The design targets 99.9% uptime (NFR-1), active-active disaster recovery with a 5-minute RTO (D-12), and server-side p95 latency under 50 ms (D-8), while staying cost-aware by treating 100,000 concurrent players as an autoscaled peak rather than a sustained load (D-3). Compliance with GDPR and COPPA is designed in from the start (SC-1, SC-2, D-14, D-18). The single most important commercial caveat is the tension between the first-year cloud budget of \$200,000 and the cost of serving ~1M DAU across multiple regions; this is addressed in Section 8 and tracked as a primary risk.

2. Scope

2.1 In scope

- Multi-region AWS cloud backend for the racing game: networking, security, compute, storage, data, and analytics (covers FR-1...FR-12, NFR-1...NFR-9, DR-1...DR-5, SC-1...SC-6).
- Real-time game-server hosting and skill/latency matchmaking on Amazon GameLift (FR-1, FR-2, D-2).
- Player identity, profiles, leaderboards, stats, and entitlements (FR-3, FR-4, FR-5, DR-1, DR-2).
- Payments integration backend: native store billing plus Stripe, storing tokens/receipts only (FR-6, D-6, D-17).

- Replay and asset storage with CDN, designed to support future replay sharing and ghost races (FR-7, DR-3).
- Analytics pipeline (engagement, monetization, balance) and real-time cheat/anomaly detection (FR-8, FR-9, FR-10, D-5, D-9).
- Secure customer-support data APIs for Zendesk integration (FR-11).
- Greenfield AWS Organization / Control Tower landing zone with dev, staging, production, and beta environments (D-10, NFR-9, FR-12).
- Infrastructure as Code in Terraform: a full multi-region production stack plus a single-region deployable demo (D-4).
- CI/CD with GitHub Actions and blue-green / per-region rollout (D-13, NFR-7).
- Observability, alerting, and operational runbooks; knowledge transfer to GC's DevOps team (NFR-8, D-16).
- Internationalization-ready architecture (languages delivered post-launch) (D-15).

2.2 Out of scope

- Unity game-client development, gameplay logic, art, and the 120 fps client target (client-side; A-4).
 - Game design, level/track design, and content authoring.
 - Translation/localization content production (architecture is provided; languages are delivered later — D-15).
 - Marketing and user-acquisition campaigns.
 - In-product machine learning / AI features; the project's "AI" is AI-assisted delivery, not an in-game ML feature (D-9, scope note in the requirements doc).
 - Procurement and contracts for third-party services (Stripe, parental-consent vendor, Zendesk, Datadog/New Relic).
 - Recruitment and staffing of the SRE team (GC hires; CA advises — D-16).
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3. Business objectives

Objective	Target / success metric	Source
Launch the first title on time	Production launch within 8 months	BR-2
Support launch-scale concurrency	>= 100,000 concurrent players (autoscaled peak)	BR-3, D-3
Differentiate on latency	Server-side p95 < 50 ms in-region; ~3 s in-race reconnect grace	D-8
High availability	>= 99.9% uptime; RTO 5 min, RPO 0 financial / <= 5 min game state	NFR-1, D-12
Global reach	Launch NA + EU; Asia within 6 months	BR-7
Rapid, low-risk releases	Weekly content + monthly feature releases, minimal downtime	NFR-7
Cost discipline	Operate toward the first-year cloud budget (see Section 8 caveat)	BR-6, D-3
Regulatory compliance	GDPR and COPPA satisfied at launch	SC-1, SC-2, D-14, D-18
Data-driven operations	KPI dashboards (DAU/MAU, ARPU, retention, match completion) and cheat detection	FR-8, FR-9, FR-10

4. Technical solution

4.1 Architecture overview

The platform is a multi-region, active-active AWS architecture. Players connect from iOS/Android clients through a global edge layer to region-local backend services and GameLift game-server fleets placed close to players to minimize latency. Shared platform services (identity, profiles, payments, analytics) are reused across GC's planned future titles (BR-1). Launch regions are one in North America and one in Europe, with an Asia-Pacific region added at expansion (BR-7); the data model is residency-aware so EU player PII stays in the EU (D-11).

Request flow at a high level:

1. The client resolves the nearest healthy region via Amazon Route 53 latency-based routing to the regional API Gateway; game-session placement across regional fleets uses GameLift FlexMatch.
2. Static content, game assets, and replays are served through Amazon CloudFront; AWS WAF and Shield protect public endpoints (SC-6).
3. Authenticated API traffic terminates at Amazon API Gateway and is handled by regional backend services (Lambda and/or containers).
4. Matchmaking (GameLift FlexMatch) groups players by skill and latency, then places sessions on GameLift fleets; the race runs on the dedicated game server.
5. Game and player state is read/written to DynamoDB and ElastiCache; gameplay/telemetry events are streamed to the analytics and cheat-detection pipelines.

4.2 Core components (AWS services)

Capability	AWS service	Notes / decision
Global routing / failover	Route 53 (latency routing)	Nearest-region API routing + active-active failover (NFR-5, D-12)
Edge / CDN	CloudFront	Assets, game updates, replay delivery (FR-7, DR-3)
Edge security	WAF, Shield	DDoS and L7 protection for public endpoints (SC-6)
Game servers	Amazon GameLift (Spot + on-demand)	Managed session hosting, autoscaling (FR-2, D-2)
Matchmaking	GameLift FlexMatch	Skill- and latency-based (FR-1)
Identity / auth	Amazon Cognito	Email/password + social; age gate + parental consent (FR-3, D-14)
API layer	API Gateway + Lambda (containers where needed)	Stateless, autoscaled backend services (BR-4)
Operational data	DynamoDB	Global tables for non-PII; region-scoped PII (FR-5, NFR-6, D-11)
Low-latency cache	ElastiCache (Redis)	Leaderboards (sorted sets), session state (FR-4)
Object storage / lake	S3	Replays (30-day lifecycle), assets, analytics lake (DR-3, DR-4)
Payments	Native store billing + Stripe	Tokens/receipts/entitlements only (FR-6, D-6, D-17)
Event streaming	Kinesis Data Streams / Firehose	Telemetry ingestion (FR-8, FR-9)
Real-time analytics	Managed Service for Flink + Lambda	Rules + statistical cheat detection (FR-9, D-9)
Warehouse / query	Athena + Glue Data Catalog	Athena now; Redshift Serverless later (FR-8, D-5)
Dashboards	QuickSight and/or Grafana	KPI and ops dashboards (FR-10)
Support APIs	API Gateway + Lambda, CloudTrail	Least-privilege Zendesk data access, audited (FR-11)
Observability	CloudWatch, X-Ray, Datadog/New Relic	APM, alerting, automated remediation (NFR-8)
Secrets / keys	Secrets Manager, KMS	Encryption and credential management (SC-3)
Landing zone	Control Tower, AWS Organizations	Multi-account, SCP guardrails (D-10, NFR-9)
CI/CD	GitHub Actions + OIDC	Blue-green, per-region rollout (D-13, NFR-7)

4.3 Multi-region and data strategy

DynamoDB is the primary operational store. Non-PII, latency-sensitive data (matchmaking metadata, leaderboards, session state) uses global tables for multi-region, multi-active access; player PII (profiles, payment references, entitlements, consent records) is region-scoped to the player's home region and is not replicated across regions, satisfying both NFR-6 ("globally distributed database") and EU residency

(SC-1, D-11). Replays and analytics data live in regional S3 buckets; the analytics lake keeps de-identified/aggregated data to limit PII spread (DR-4, D-18).

4.4 Scaling and cost optimization

100,000 concurrent players is treated as an autoscaled peak over a low baseline, not a sustained load (D-3). GameLift fleets use Spot capacity with on-demand fallback and scale toward zero in off-peak windows; DynamoDB uses on-demand or autoscaling capacity; backend services are stateless and horizontally scaled (BR-4, NFR-4). Autoscaling limits and quotas are sized to burst to 5x the 100,000 peak for seasonal/viral events (D-12). Cost guardrails include budget alarms, Savings Plans for steady baseline, and S3 lifecycle/tiering for replays and cold analytics (BR-6).

4.5 Environments, CI/CD, and operations

A greenfield AWS Organization with a Control Tower landing zone provides separate dev, staging, production, and beta accounts with baseline SCP guardrails (D-10, NFR-9). The beta environment mirrors production at limited capacity for the 10,000-tester program (FR-12). CI/CD runs on GitHub Actions authenticating to AWS via OIDC (no static keys), using blue-green deployments and gradual per-region rollout during low-traffic windows, with automated rollback (D-13, NFR-7).

4.6 Analytics and anti-cheat

Game-client and server events are ingested through Kinesis into the S3 data lake for batch/interactive analysis with Athena and Glue, feeding KPI dashboards (FR-8, FR-10, D-5). A parallel real-time path (Kinesis + Managed Service for Flink / Lambda) applies rules and statistical anomaly detection to flag cheating and abnormal behavior at launch, with the pipeline designed so an ML model can be added later without rework (FR-9, D-9).

5. Security and compliance

5.1 Identity and access

Amazon Cognito handles player authentication (email/password and social providers, configurable). A neutral age gate runs at signup; under-13 players require third-party verifiable parental consent (D-14, SC-2). Internal and support APIs enforce least-privilege IAM, and all access to player data is audited via CloudTrail and application access logs (FR-11, SC-5).

5.2 Encryption and secrets

All data is encrypted at rest with AWS KMS and in transit with TLS 1.3 (SC-3). Credentials and API keys are stored in AWS Secrets Manager. No static cloud credentials are used in CI/CD (OIDC, D-13).

5.3 GDPR

EU player PII is pinned to the EU region (D-11, SC-1). Data-subject export and erasure requests are honored without undue delay and within one month of receipt, extendable by two further months for complex requests with notice in the first month (D-18, verified against current GDPR Art. 12(3) / Art. 17). Erasure removes data from active stores immediately, de-identifies analytics data, and lets backups age out on their retention cycle with deletions re-applied on restore. A data map maintains coverage across all stores.

5.4 COPPA

Age verification and verifiable parental consent gate under-13 accounts; data collection and processing for minors is restricted, and consent records are retained (SC-2, D-14).

5.5 Payments and PCI

Monetization is a one-time game purchase plus in-app purchases for cosmetic add-ons; no subscription (D-17). Payments are processed through native store billing and Stripe; the platform stores only receipts, entitlements, and PSP tokens — never raw card data — keeping it out of full PCI-DSS scope (FR-6, SC-4, D-6).

5.6 Network protection and account guardrails

Public endpoints are fronted by WAF and Shield (SC-6). The Control Tower landing zone applies baseline Service Control Policies across the organization (D-10). Recommended baseline SCPs:

- Deny public S3 bucket access (block public ACLs/policies org-wide).
- Require encryption for S3 and EBS; deny creation of unencrypted resources.
- Require IMDSv2 on EC2 / GameLift fleet instances.
- Deny root-account access keys and enforce MFA for privileged roles.
- Restrict resource creation to approved regions (NA, EU, and APAC at expansion).
- Deny disabling of CloudTrail, AWS Config, and GuardDuty.

GC does not currently have an organization, so these are delivered as part of the greenfield landing zone (D-10).

6. Timeline and milestones

Eight months, phased, with critical-path items (DevOps onboarding, GameLift/Unity integration, compliance) started early. Durations are indicative and run with parallel workstreams.

Phase	Duration	Key outputs
1. Discovery and architecture	Weeks 1-4	Requirements (done), architecture + diagram, landing zone design, security/compliance plan
2. Foundation	Weeks 4-8	Control Tower org, networking, Terraform modules, CI/CD, observability baseline
3. Core backend	Weeks 8-20	Auth/profiles, matchmaking + GameLift fleets, leaderboards, payments backend, analytics pipeline
4. Integration and hardening	Weeks 16-24	Client/backend integration, multi-region active-active, DR drills, security testing
5. Testing and beta	Weeks 24-30	Load test to 100k CCU, 10,000-player beta, performance tuning, compliance validation
6. Launch readiness and go-live	Weeks 30-34	Production cutover, blue-green rollout, launch-readiness sign-off, runbooks/training
Post-launch	Weeks 34+	Stabilization, Asia-region expansion prep, KPI review

Key milestones: architecture sign-off (end of Phase 1), foundation/landing zone ready (end of Phase 2), feature-complete backend (end of Phase 3), 100k-CCU load test passed and beta complete (end of Phase 5), compliance validated and go-live (Phase 6).

7. Risks and mitigations

#	Risk	Impact	Mitigation
R-1	First-year budget vs ~1M DAU multi-region cost (BR-6, D-3)	Budget overrun	Cost-optimized peak, Spot, scale-to-low, S3 lifecycle/tiering, budget alarms; reforecast early and escalate budget if needed (Section 8)
R-2	p95 < 50 ms aggressive on mobile last-mile (D-8)	Missed latency target	Latency-based matchmaking caps match RTT; more regional points of presence; define as server-side RTT; validate under load and relax if needed
R-3	Team new to cloud operations	Delivery/ops risk	Managed services, IaC, runbooks, training, and CA guidance; phased knowledge transfer
R-4	SRE hiring timeline (D-16)	24/7 readiness at launch	Phased on-call, interim CA guidance, automated remediation; track hiring against launch date
R-5	GameLift Spot interruptions	Match disruption	On-demand fallback, baseline reserved capacity, session resilience and reconnect grace (NFR-2)
R-6	GDPR/COPPA gaps for global minors	Legal/financial	Early legal engagement, consent vendor (D-14), data map and erasure SLA (D-18), audits
R-7	8-month timeline is aggressive (BR-2)	Schedule slip	Phased MVP, parallel workstreams, managed services, early critical-path starts
R-8	5x burst-event cost and cross-region transfer (D-12)	Cost spikes	Event-scoped autoscaling, traffic locality, monitor cross-AZ/region transfer; bursts are demand/revenue-driven

8. Cost and budget

The first-year cloud-infrastructure budget is \$200,000, i.e. roughly \$16,700 per month on average (BR-6). The figures below are rough order-of-magnitude monthly ranges for planning only; a detailed model using AWS pricing data is a design-phase deliverable.

Cost area	Low baseline (off-peak)	Notes
Game compute (GameLift, Spot-heavy)	4,000 - 12,000	Scales toward zero off-peak; spikes during events
DynamoDB (global + region-scoped)	2,000 - 5,000	On-demand/autoscaling
ElastiCache (leaderboards/session)	1,000 - 3,000	
S3 storage (replays ~225 TB rolling + assets + lake)	5,000 - 8,000	Major driver; optimize via tiering, compression, retention (DR-3)
CloudFront + data transfer	2,000 - 5,000	Includes cross-region transfer
Streaming + analytics (Kinesis/Flink/Athena/Glue)	1,000 - 3,000	
Edge security, Cognito, API/Lambda, observability	1,000 - 3,000	WAF/Shield, CloudWatch, etc.
Indicative monthly total (low baseline)	~16,000 - 39,000	Multi-region overhead adds ~20-40%

Finding: at ~1M DAU (D-7) across multiple regions, even a cost-optimized design tends toward or above the \$200,000/year ceiling at the low end, and well above it during 5x events. Replay storage is a notable single driver. Recommendation: treat \$200,000 as a baseline-only target and either (a) increase the cloud budget in line with realized DAU, (b) reduce replay retention or apply compression/tiering, or (c) accept higher, demand-driven spend during growth periods. This is tracked as risk R-1 and was flagged in D-3.

9. Roles, operations, and support

Operations follow the client-staffed model: GC recruits the three shift SREs while CA provides runbooks, training, on-call/automation tooling design, and architecture guidance alongside GC's two incoming DevOps engineers (D-16, NFR-8). Observability uses CloudWatch and X-Ray with Datadog or New Relic for unified APM, plus automated remediation for common faults.

Indicative support tiers (refined during operational design):

Tier	Scope	Target initial response
P1 critical	Outage, payment failure, security incident	15 minutes, 24/7
P2 high	Matchmaking failures, severe latency, auth problems	1 hour, 24/7
P3 standard	Non-blocking bugs, individual account issues	4 business hours
P4 maintenance	Scheduled changes, optimizations	Next business day

10. Deliverables

- This SOW (/sow/SOW.md plus PDF export).
- Architecture design and diagram (/architecture/diagram.png).

- Terraform code (`/terraform/`): full multi-region production stack and a single-region deployable demo, organized into modules (networking, security, compute, storage, data, etc.) (D-4).
- Validation evidence (endpoint availability, storage access, baseline security checks).
- `README.md` with setup and deployment instructions.
- Operational runbooks and knowledge-transfer materials (NFR-8, D-16).

The single-region demo is deployed live (us-east-1) via a least-privilege IAM deployer with S3-locked remote state (D-19, D-20). It mirrors the production data plane and edge as closely as is deployable; three components are documented but not instantiated (GameLift, Managed Flink, QuickSight). The full command-by-command run and a resource inventory are in [04-deployment-log.md](#); demo scope and deltas are in architecture Section 19.

11. Assumptions and traceability

This SOW inherits the assumptions recorded in the requirements document (Section 8 there), notably:

- auth methods (email/password plus social) are design-ready and to be confirmed (A-7).
- the \$200,000 figure covers cloud infrastructure only (A-8).
- the studio is 15 developers plus 2 incoming DevOps engineers (A-9).

Full requirement-to-decision traceability (BR/FR/NFR/DR/SC and D-1...D-18) lives in [01-requirements-and-architectural-implications.md](#), which is the controlling document if any detail here and there diverge.